

INFO8006 Introduction to Artificial Intelligence

Exam of January 2026

Instructions

- The exam lasts for 4 hours.
- You are allowed to use a calculator during the exam, but documents of any kind are forbidden.
- The last two pages can be used for scratch work or for extra space. If you want work done there to be graded, mention where to look **in big letters with a box around them**, on the page with the question.
- Write your last name, first name, and ULiège ID on the first page. Write only your ULiège ID on all the other pages.
- Before handing in your exam, **sort all the pages according to the page numbers** (even if you used additional pages to answer a question).

Good luck!

LAST NAME, FIRST NAME (in uppercase): _____

ULiège ID (s201234): _____

Question 1 [4 points] Multiple choice questions. Choose one of the four choices by filling in its circle. Correct answers are graded $+\frac{4}{10}$, wrong answers are graded $-\frac{2}{15}$ and the absence of answers is graded 0. The total of your grade for Question 1 is bounded below at 0/4.

1. A utility-based agent differs from a goal-based agent in that ...
 - ☐ it uses condition-action rules.
 - ☐ it can handle partially observable environments.
 - ☐ it does not need a model of the environment.
 - ☐ it selects actions based on a performance measure over state sequences.
2. If a heuristic h is admissible but not consistent, then A^* with graph search ...
 - ☐ becomes equivalent to breadth-first search.
 - ☐ is guaranteed to find the optimal solution.
 - ☐ may fail to find the optimal solution.
 - ☐ will consistently have a higher time complexity than A^* with tree search.
3. Alpha-beta pruning is most effective when ...
 - ☐ the best moves for each player are explored first.
 - ☐ the evaluation function returns only positive values.
 - ☐ the worst moves for each player are explored first.
 - ☐ the tree is perfectly balanced with equal branching factors.
4. For events A and B with $P(B) > 0$, we always have ...
 - ☐ $P(A|B) = P(B|A)$.
 - ☐ $P(A|B) + P(\neg A|B) = 1$.
 - ☐ $P(A|B) \leq P(A)$.
 - ☐ $P(A|B) + P(\neg A|B) = P(B)$.
5. Adding edges to a Bayesian network ...
 - ☐ may remove conditional independence assumptions.
 - ☐ always improves the accuracy of probabilistic inference.
 - ☐ reduces the number of parameters needed.
 - ☐ may introduce new conditional independencies.
6. The Viterbi algorithm computes ...
 - ☐ the expected reward of following the optimal policy.
 - ☐ the probability distribution over the current hidden state.
 - ☐ the probability of the observed evidence sequence.
 - ☐ the most likely sequence of hidden states given the observations.
7. The cross-entropy loss for classification ...
 - ☐ is only applicable to binary classification problems.
 - ☐ measures the Euclidean distance between predictions and targets.
 - ☐ is derived from the maximum likelihood principle for categorical distributions.
 - ☐ requires a ReLU activation function in the output layer.
8. Max-pooling layers in convolutional neural networks ...
 - ☐ are required before every convolutional layer.
 - ☐ reduce spatial dimensions while preserving dominant features.
 - ☐ increase the spatial resolution of feature maps.
 - ☐ introduce learnable parameters for downsampling.

9. In an MDP, increasing the discount factor γ towards 1 ...
- ☐ violates the Markov property.
 - ☐ makes the agent more short-sighted.
 - ☐ guarantees that the optimal policy changes.
 - ☐ may slow down the convergence of value iteration.
10. Q-learning is called an off-policy algorithm because ...
- ☐ it can learn the optimal policy while following a different behavior policy.
 - ☐ it requires the transition probabilities to be known.
 - ☐ it learns a model of the environment.
 - ☐ it does not use temporal-difference updates.

Question 2 [4 points] A car navigates an $M \times N$ grid to reach a designated parking spot. The state of the car is described by its position (x, y) , its direction $d \in \{N, S, E, W\}$, and its velocity $v \in \{0, 1, \dots, V_{max}\}$. At each time step, the car moves v cells forward in direction d . Actions allow the car to adjust its direction and velocity. The goal is to reach position (x_{goal}, y_{goal}) with velocity $v = 0$.

- (a) If the grid is $M \times N$, what is the size of the state space? Explain your answer.
- (b) Is the Manhattan distance a good heuristic function for this search problem? Explain your answer.

- (c) If we used A^* *graph* search with an inadmissible heuristic, would the search be complete? Would it be optimal? Explain your answers.

- (d) If we used A^* *graph* search with an admissible heuristic, would the search be optimal? If yes, prove it formally. If not, explain why not and provide a solution that would make it optimal.

Question 3 [4 points] A meteorological station monitors weather conditions. The true temperature T (in °C) at its location is modeled as a Gaussian random variable with prior distribution $T \sim \mathcal{N}(15, 9)$ (mean 15°C, variance 9).

Two sensors provide noisy measurements of the temperature:

- Sensor 1 gives reading $S_1|T \sim \mathcal{N}(T, 4)$,
- Sensor 2 gives reading $S_2|T \sim \mathcal{N}(T, 1)$.

Additionally, humidity H (in %) depends on temperature according to $H|T \sim \mathcal{N}(2T + 30, 16)$.

- (a) Draw the Bayesian network corresponding to this model. Specify all nodes and their associated probability distributions.

- (b) Answer the following independence questions. Justify your answers using d-separation.

- (i) Are S_1 and S_2 marginally independent? ($S_1 \perp S_2$?)
- (ii) Are S_1 and S_2 conditionally independent given T ? ($S_1 \perp S_2|T$?)
- (iii) Are S_1 and H conditionally independent given S_2 ? ($S_1 \perp H|S_2$?)

(c) Using the Gaussian identities in Appendix A, derive the posterior distribution $p(T = t|S_1 = 18)$.

(d) Now suppose both sensors give readings: $S_1 = 18$ and $S_2 = 16$. Derive the posterior distribution $p(T = t|S_1 = 18, S_2 = 16)$.

(e) Derive the predictive distribution $p(H = h|S_1 = 18)$.

Question 4 [4 points] In intensive care, bacterial infections are a major concern. Nurses perform a rapid CRP test on patients every few hours in order to decide whether or not to administer an antibiotic. However, the test is noisy: it returns a positive result in 90% of cases if the patient is infected, and a negative result in 70% of cases if the patient is healthy.

We model the state of a patient at time t as $X_t \in \{\text{healthy}, \text{infected}\}$ and the test result as evidence $E_t \in \{+, -\}$. At $t = 0$, a patient is at home with uniform probability of being healthy or infected. The patient then arrives at the hospital and spends some time in the waiting room before being examined. During this time, no test is performed and no antibiotic is administered. The dynamics without antibiotic are as follows: an infected patient remains infected with certainty, while a healthy patient has a 20% chance of becoming infected. Monitoring begins at $t = 1$.

- (a) Define the prior distribution $\mathbf{P}(X_0)$ and the transition model $\mathbf{P}(X_1|X_0)$. Derive the predicted distribution $\mathbf{P}(X_1)$ before any observation.

- (b) Define the observation model $\mathbf{P}(E_t|X_t)$. The first CRP test at $t = 1$ returns negative. Derive $\mathbf{P}(X_1|e_1 = -)$.

- (c) From $t = 1$ onwards, the nurse decides on an action $a_t \in \{\text{antibiotic}, \text{no antibiotic}\}$ after observing the test result e_t . The action affects the transition from X_t to X_{t+1} :
 - If an antibiotic is administered ($a_t = \text{antibiotic}$): an infected patient has a 60% chance of becoming healthy, while a healthy patient has a 10% chance of becoming infected (due to drug-resistant bacteria).
 - If no antibiotic is administered ($a_t = \text{no antibiotic}$): dynamics are as described above.

Draw the (dynamic) Bayesian network representing the model over the time steps $t = 0, 1, 2, 3$.

- (d) Define the transition model $\mathbf{P}(X_{t+1}|X_t, a_t)$. Derive the general filtering recursion for $\mathbf{P}(X_t|e_{1:t}, a_{1:t-1})$ in terms of the model components.
- (e) Starting from $\mathbf{P}(X_1|e_1)$ computed in part (b), compute the sequence of belief states $\mathbf{P}(X_t|e_{1:t}, a_{1:t-1})$ for $t = 2, 3$ given the evidence-action pairs $(e_1, a_1) = (-, \text{no antibiotic})$, $(e_2, a_2) = (+, \text{antibiotic})$, and $(e_3, a_3) = (-, \text{antibiotic})$.
- (f) After some thought, the nurse wonders whether the patient might already have been infected when the test returned negative at $t = 1$. He wants to compute the smoothing distribution $\mathbf{P}(X_1|e_{1:3}, a_{1:2})$. Discuss whether $a_1 = \text{no antibiotic}$ was a good decision in hindsight, based on the smoothing result.

Question 5 [4 points] We consider a neural network for binary classification that takes as input a 2D point $(x_1, x_2) \in \mathbb{R}^2$ and outputs a probability $\hat{y} = f(x_1, x_2) \in [0, 1]$. The network has one hidden layer with two units and is defined as

$$\begin{aligned} h_1 &= \text{ReLU}(x_1 - 1) \\ h_2 &= \text{ReLU}(x_2 - 1) \\ \hat{y} &= \sigma(h_1 + h_2 - 1), \end{aligned}$$

where $\text{ReLU}(z) = \max(0, z)$ is the rectified linear unit and $\sigma(z) = \frac{1}{1+e^{-z}}$ is the sigmoid function. The predicted class is $\hat{c} = 1$ if $\hat{y} \geq 0.5$ and $\hat{c} = 0$ otherwise.

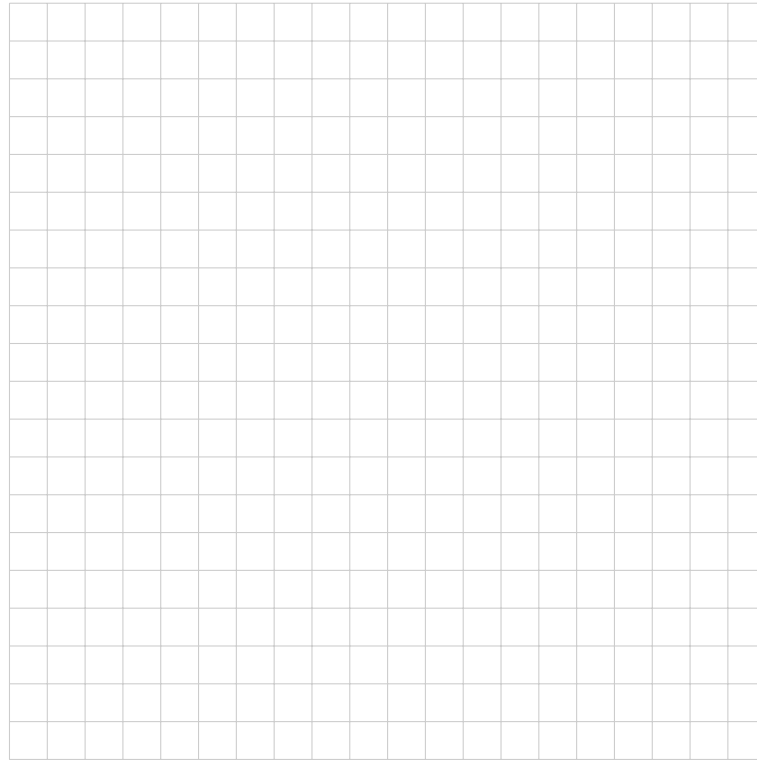
- (a) Draw the computation graph representing the neural network. Use circled nodes for variables (inputs, outputs, intermediate values) and squared nodes for operations (addition, ReLU, sigmoid).

- (b) The ReLU activation functions divide the input space into several regions, depending on whether each hidden unit is active ($h_i > 0$) or inactive ($h_i = 0$).

(i) Identify and characterize these regions in terms of conditions on x_1 and x_2 .

- (ii) For each region, give a simplified expression for the output probability $\hat{y}$ as a function of x_1 and x_2 .

- (c) Plot the decision boundary $\hat{y} = 0.5$ of the classifier on the grid below. Clearly indicate which region(s) are classified as class 1 and which as class 0.



- (d) We now consider a modified network with a third hidden unit:

$$h_1 = \text{ReLU}(x_1 - 1)$$

$$h_2 = \text{ReLU}(x_2 - 1)$$

$$h_3 = \text{ReLU}(x_1 + x_2 - 3)$$

$$\hat{y} = \sigma(h_1 + h_2 - h_3 - 1)$$

On the grid above and using a different color, plot the new decision boundary and compare it to the original one. Explain what is the effect of adding h_3 ?

Appendix A. Gaussian identities (Särkkä, 2013).

(a) If $\mathbf{x}$ and $\mathbf{y}$ have the joint Gaussian distribution

$$p\left(\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix}\right) = \mathcal{N}\left(\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} \middle| \begin{pmatrix} \mathbf{a} \\ \mathbf{b} \end{pmatrix}, \begin{pmatrix} \mathbf{A} & \mathbf{C} \\ \mathbf{C}^T & \mathbf{B} \end{pmatrix}\right),$$

then the marginal and conditional distributions of $\mathbf{x}$ and $\mathbf{y}$ are given by

$$\begin{aligned} p(\mathbf{x}) &= \mathcal{N}(\mathbf{x}|\mathbf{a}, \mathbf{A}) \\ p(\mathbf{y}) &= \mathcal{N}(\mathbf{y}|\mathbf{b}, \mathbf{B}) \\ p(\mathbf{x}|\mathbf{y}) &= \mathcal{N}(\mathbf{x}|\mathbf{a} + \mathbf{CB}^{-1}(\mathbf{y} - \mathbf{b}), \mathbf{A} - \mathbf{CB}^{-1}\mathbf{C}^T) \\ p(\mathbf{y}|\mathbf{x}) &= \mathcal{N}(\mathbf{y}|\mathbf{b} + \mathbf{C}^T\mathbf{A}^{-1}(\mathbf{x} - \mathbf{a}), \mathbf{B} - \mathbf{C}^T\mathbf{A}^{-1}\mathbf{C}). \end{aligned}$$

(b) If the random variables $\mathbf{x}$ and $\mathbf{y}$ have Gaussian probability distributions

$$\begin{aligned} p(\mathbf{x}) &= \mathcal{N}(\mathbf{x}|\mathbf{m}, \mathbf{P}) \\ p(\mathbf{y}|\mathbf{x}) &= \mathcal{N}(\mathbf{y}|\mathbf{H}\mathbf{x} + \mathbf{u}, \mathbf{R}), \end{aligned}$$

then the joint distribution of $\mathbf{x}$ and $\mathbf{y}$ is Gaussian with

$$p\left(\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix}\right) = \mathcal{N}\left(\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} \middle| \begin{pmatrix} \mathbf{m} \\ \mathbf{H}\mathbf{m} + \mathbf{u} \end{pmatrix}, \begin{pmatrix} \mathbf{P} & \mathbf{P}\mathbf{H}^T \\ \mathbf{H}\mathbf{P} & \mathbf{H}\mathbf{P}\mathbf{H}^T + \mathbf{R} \end{pmatrix}\right).$$

Extra page 1 / 2.

Extra page 2 / 2.